Romans and The Arabs
400 BC -1300 AC

Before

Civilizations before the Greeks developed agriculture, practiced medicine, discovered numbers, math and applied engineering and used calendars (lunar, solar).

Greeks used science not only as a practical tool but also to explain nature. Greeks wanted strictly deductive proofs based on stated axioms

Romans

Roman civil engineering and military engineering constituted a large part of Rome's technological superiority and legacy.

The construction of a vast and efficient travel network throughout the Empire dramatically increased Rome's power and influence.

Romans

It was originally constructed to allow Roman legions to be rapidly deployed.

But these highways also had enormous economic significance, solidifying Rome's role as a trading crossroads - the origin of the saying "all roads lead to Rome".

Romans – Legal System

Most early civilizations were ruled by custom or the arbitrary judgments of kings or priests

Laws and the punishments for not following the laws were at the whim of the ruler.

In 450 BC, after a revolt of plebs who felt they were entitled to know and be able to interpret the code of laws, the Twelve Tables were established.

http://www.historyguide.org/ancient/12tables.html

Romans – 12 Tables

TABLE I Procedure: for courts and trials

TABLE II Trials continued & Theft

TABLE III Debt

TABLE IV Rights of fathers (paterfamilias) over the family

TABLE V Legal guardianship and inheritance laws

TABLE VI Acquisition and possession

TABLE VII Land rights & crimes

TABLE VIII Torts and delicts (Laws of injury)

TABLE IX Public law

TABLE X Sacred law

TABLE XI Supplement I

TABLE XII Supplement II

Romans – Modern Legal Notions

<u>Concepts of</u>

- Lawyer - The Venice Story
- Jury to give verdict
- Prison to keep person under arrest until trial

Corpus Juris included modern legal concepts such as

"the accused is innocent until proven guilty."

The judgment and punishments are based on precedent, on rulings that previous judges and juries have made.

Roman – Modern Government

Division of power

The contemporary idea of democracy, while borne out of the political struggles in the city of Athens, came to fruition in the Roman Republic. From Monarchy to Representation

Instead of a king there are 3 parties to guards against despotism:

- **Consuls**
- **Senate**
- **Assemblies**

Roman – Modern Government

Division of power

Consuls: The new government chose consuls, two in number. These individuals were not elected by the populace but appointed by the popular assembly.

They are both political and military heads of state, consuls possessed supreme executive power, commanding the army, presiding over the Senate, and proposing legislation; however, as a safeguard each consul had the ability to veto the other's decision

Roman – Modern Government

<u>Senate;</u> while it had no legal power, it still held significant influence, serving as an advisory body to the consuls and later emperors. Members of this conservative body were unpaid and served for life unless found guilty of public or private misconduct. Senators were forbidden to engage in banking or foreign trade. *Conflict of Interest.*

<u>Assemblies;</u> has power to pass law. They are representatives of tribes/people to some extend.

Roman – Government

The Roman government of the old Republic had created a unique system of the division of power which was a safeguard against oppression by any single individual.

Power, for the most part, lay with a voting public.

There were elected officials and a representative body.

Under the emperor, the government changed, and the emperor became the ultimate decision maker, he had the last word.

Rome must, nevertheless, remain an outstanding example of a successful ancient government.

10 Innovations That Built Ancient Rome

http://www.history.com/news/history-lists/10-innovations-that-built-ancient-rome

1. Aqueducts
2. Concrete
3. Newspaper
4. Welfare
5. Bound Books
6. Roads and Highways
7. Roman Arches
8. Julian Calendar
9. The Twelve Tables and the Corpus Juris Civilis
10. Battlefield Surgery

Calendar

Lunar calendar: Circle is divided to 12 months of 30 days each. (Hence 360 degrees)

Julian (Julius Caesar) calendar: 365 + ¼ days in a year. However, this caused shift in seasons.

Because Earth revolves around 365.2425 days

Gregorian (modern) calendar: Corrects Julian calendar by skipping the leap year by every 100 years and keeping it in every 400 years.

Roman Months

30, 29, 30, 29, 30, 29, 30, 29, 30, 29, 30, 29

355 days – Corrected by adding month in intercalary year. Which is decided by politics.

31, 29/30, 31, 30, 31, 30, 31, 30, 31, 30, 31, 30

Julius Ceaser added 1 day + leap year

Agustus modified as below.

31, 28/29, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31

Ianuarius

Februarius

Martius

Aprilis

Maius

Iunius

Quintilis → Julius

Sextilis → Augustus

September

October

November

December

Fall of Romans – 500th

The arrival of the Huns and the migration of the Barbarian tribes

Over expansion and military overspending

Government corruption causing political instability

Christianity and the loss of traditional values

Economic troubles and over reliance on slave labor

Rise of Arabs and Islamic Scholars

Western civilization collapsed. A lot of work is lost. Fragments of Greek work kept/translated by Arabs, and they further developed

- Aristotle's work is 12th century Arab translation
- Ptolemy's work is 9th century

In Islam. «Read» the first order.

There was the blief that «Learn knowledge, even it is in China»

Abasis promoted Science and invested…

Progress under Arabs

Arabs developed Algebra, alchemy, algorithm, average, almanac, aorta, and alcohol (Arabic origin words)

Finding direction to Mecca and Pray time required accurate measurement of time/location due to Sphere Shape of Earth – establishment of observatories and advances in cartography

Hints of galaxies in Al-Sufi and Al-Burini (Milky way, 986)

Accurate Prediction of Eclipses

Al-Khwarizmi - Algebra

One of the first Directors of the **House of Wisdom** in Bagdad.

He oversaw the translation of the major Greek and Indian mathematical and astronomy works (including those of Brahmagupta) into Arabic. He promoted the Hindu Numerical system. 1-9 and 0.

The word "**algorithm**" is derived from the Latinization of his name, and the word "**algebra**" is derived from the Latinization of "**al-jabr**", part of the title of his most famous book, in which he introduced the fundamental algebraic methods and techniques for solving equations.

Progress under Arabs

Omer Khayyam measured the length of year as 365.24219858156 day. (2/100 million error).

Ibni Sina adapted the medical traditions of Galen and thereby Hippocrates. He sought to fit these traditions into Aristotle's natural philosophy, 1025. The result was a "clear and ordered "summa" of all the medical knowledge of Ibn Sinas time in *The Canon of Medicine*.

Progress in Science - Arabs

Improved glass for optics – Theory of Refraction was developed

Quarantine (Arabic) to control spread of diseases

Surgery of Eye, Ear Throat.

Blood circulation was described Al-Nafis

Jabir may be the father of Chemistry – Experiments in Labs

Al-Kwarizmi (Baghdad) wrote book Arithmetic – Algebra.

Paper was developed. Enabled publication of Books.

- Establishment of Library – The house of wisdom in Baghdad

Optics

Ibn al-Haytham most famous work is his seven-volume treatise on optics *Kitab al-Manazir* (*Book of Optics*), written from 1011 to 1021.

- Reflections, Mirrors, Perspective
- How we see objects by Eye (Intromission theory)

Ibn Saul Discovers the Law of Refraction

- Bending of Light in a Medium, Snell's Law

Surgery

Al-Zahrawi devoted his entire life and genius to the advancement of medicine as a whole and surgery.

Ottomans – 1400 - 1900

Unfortunately, Limited Contribution

The Samarkand school of geography and astronomy provided the primary sources for the Ottoman's knowledge of geography, map-making and related fields.

Contributions in Astronomy and Map making

http://www.muslimheritage.com/article/ottoman-contributions-science-and-technology

Ali Kuşçu - 1403

Kuşçu improved on Nasir al-Din al-Tusi's planetary model and rejected Aristotelian physics and completely separated natural philosophy from Islamic astronomy, allowing astronomy to become a purely empirical and mathematical science.

He found empirical evidence for the Earth's rotation through his observation on comets and concluded, on the basis of empirical evidence rather than speculative philosophy, that the moving Earth theory is just as likely to be true as the stationary Earth theory.

Ottomans – Astronomy

Kuşcu (Qushji's) view on the Earth's motion was like the later views of Nicolaus Copernicus on this issue, though it is uncertain whether the former had any influence on the latter.

Chief astronomer Taqī al-Dīn al-Rasid Worked at Galata then established Tophane

Sextant (rules) to measure distance between stars

Measured Equinoxes

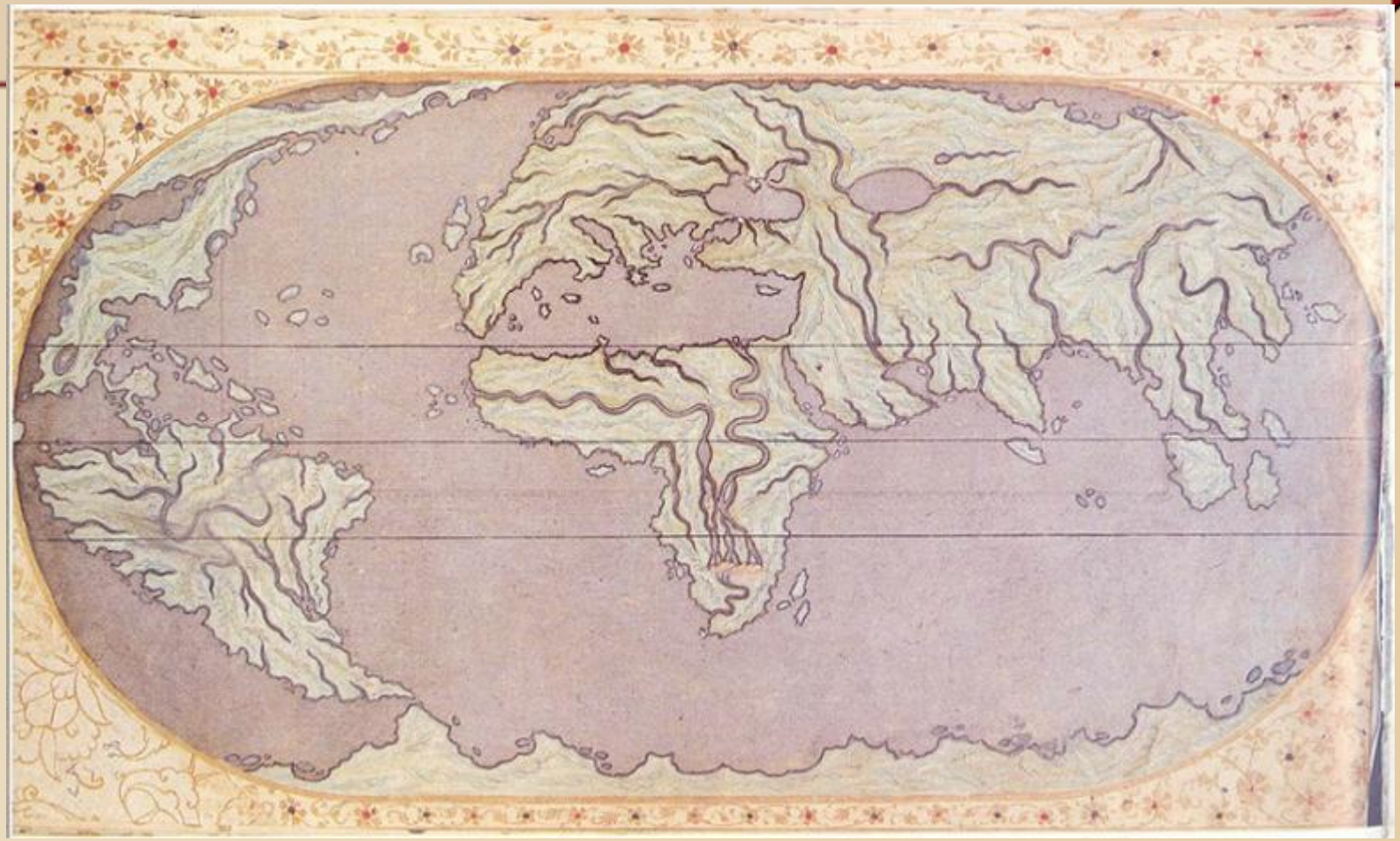
Astronomical Mechanical Clock

Ottomans – Cartography

Piri Reis completed a map of the world in 1513 and updated in 1523. It was drawn based on both his detailed drafts and European maps, including Columbus' map of America. The map does not contain latitudinal and longitudinal lines but includes coastlines and islands and serves to familiarize one with the various regions of the world.

He wrote Kitāb-i Bahriye.

World Map Based on Western Sources



Muhammad b. Amir al-Suûdī al-Niksari presented Sultan Murad III World Map, 1573.

Ottomans know not only East but Western World.

Mimar Sinan - 15th Century

Sinan born from Greek parents, great architect and has been compared to Michelangelo of the European Renaissance.

He built hundreds of buildings including mosques, palaces, harems, chapels, tombs, schools, almshouses, madrassahs, caravan serais, granaries, fountains, aqueducts and hospitals.

His inventive experimentation with centralized domed spaces, often compared with parallel developments in Renaissance Italy,

He produced monuments in which the central dome appeared weightless and the interior surfaces bathed in light.

What the Ancients Knew - Rome

More than 2,000 years ago, the Roman army swept across Europe, northern Africa, and the Middle East. Backed by the legions, their military, and engineering skills, the Romans built one of the largest empires in history, dominating as many as 36 modern nations.

https://www.youtube.com/watch?v=BNbGjI-MEXE